

# MILES FEHELEY



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Burlington, VT



[miles-feheley](#)

## EDUCATION

**University of Vermont** *Expected Graduation - May 2026*

B.S. in Mechanical Engineering, minor in Electrical Engineering  
GPA: 3.43/4.0

Relevant Coursework: Computational Fluid Mechanics, Compressible Flow, Dynamics, Computational Solids Engineering, Applied Thermodynamics, Materials Engineering, Electronics, Circuits, Digital Control With Embedded Systems, Heat Transfer

## SKILLS

- 3D Printing
- Laser Cutting
- CNC Milling
- FEA Analysis
- Leadership
- Communication
- Public Speaking

## SOFTWARES

- Solidworks
- AutoCAD
- Autodesk Inventor
- Python
- MATLAB
- ABAQUS
- ANSYS
- 3D Slicing Software
- Arduino
- Excel

## CERTIFICATIONS

- ABAQUS FEA Essentials Certificate - 2025
- EMT Certification, New Jersey Emergency Medical Services - 2021, 2023 Recertification

## PROFESSIONAL EXPERIENCE

### UVM FabLab

**Burlington, VT**

*Operations Manager*

January 2024 - Present

- Led and organized a team of fifteen people to ensure operations were mainstreamed, communication was efficient, and output was high quality.
- Integrated new technology and material types, as well as increasing lab sustainability by implementing filament recycling systems.
- Worked directly with University Researchers and Professors to optimize print quality and output speeds.
- Directed lab purchasing to meet budget and ensure sufficient material for all projects.
- Ran regular chemical and mechanical safety inspections on all stations in the lab.
- Directed training of new staff to ensure competence and effectiveness in the lab.

*Lab Technician*

March 2023 - December 2023

- Managed and maintained production of over 1500 3D-Prints and Laser Cuts.
- Developed proficiency in 3D model optimization and operating slicing programs such as Bambu Studio, PrusaSlicer, GrabCad, PreForm, and Cura.
- Completed frequent preventative maintenance as well as troubleshooting issues on assigned machines.
- Worked with students to provide post-processing options depending on project needs.

### Glen Ridge Volunteer Ambulance Squad

**Glen Ridge, NJ**

*Emergency Medical Technician*

March 2020 - June 2025

- Delivered emergency medical services to over 300 patients in a wide range of medical emergencies.
- Established efficient and effective methods of communicating with colleagues as well as other emergency services

## RELEVANT PROJECTS

### Inductive Pressure Sensor System

Collins Aerospace, 2025

Designed an inductive pressure sensor system, including the sensor itself, housing, signal processing, and display. Worked as the lead Mechanical designer and Financial manager. This project was completed as a part of a Capstone Program at UVM, working directly with Collins Aerospace. The sensor system was designed to accurately measure the pressure in a fuel tank with an MTBF of under 250,000 hours. The purpose of this sensor is to be integrated into new aircraft beginning in the next 2 years.

*Developed Skills - Signal Processing, Microcontroller Design, PCB Design, Aerospace Design*

### Microcontroller Guided Robotic Arm

UVM, May 2025

In this solo project, I designed, fabricated, and programmed a three-axis articulatable robotic arm. The arm was comprised of various 3D printed and laser cut components, and was programmed using an Arduino Mega 2560, controlled using multiple stepper and servo motors in conjunction with a set of joysticks to steer. The finished product was able to lift a coffee cup off a table and move it to a new location without spilling.

*Developed Skills - Arduino, Power Management, Programming, Fabrication, Troubleshooting*